

FORMATTING AND PARTITIONING AN .IMG / .BIN IMAGE FILE (LINUX)

OVERVIEW TO FORMAT WITH A SINGLE FILESYSTEM (SINGLE PARTITION)

*(WILL OVERWRITE & WIPE ANY DATA ON TARGET IMAGE FILE & FILESYSTEMS)

- 1- Ensure the target .img / .bin image file is not in use / not attached as a loop device.
- 2- erase filesystem, raid or partition-table signatures (magic strings) from the specified *device* USING: wipefs --force -a <target image file>
- 3- Define a partition label type (gpt or mbr) to the target loop device. USING: parted -s <target image file> mklabel <gpt or mbr>
- 4- Create a single (1st) partition with desired fs-type spanning entire target device. USING: parted -s -a opt <target image file> mkpart <fs-type> 0% 100%
- 5- Mount target image file as a loop device. USING: losetup -fP <target image file>
- 6- Format the target loop device partition with desired fs-type. USING : mkfs command

[EXT4]: mkfs.ext4 -F -L <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>

[BTRFS]: mkfs.btrfs -f -L <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>

[FAT32]: mkfs.fat -F 32 -n <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>

[exFAT]: mkfs.exfat -n <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>

[NTFS]:
mkfs.ntfs --quick --force --label <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>
- 7- Detach the loop device associated with the target image file. USING: losetup -d <loop dev>

OVERVIEW TO FORMAT WITH MULTIPLE FILESYSTEMS (MULTIPLE PARTITIONS)

*(WILL OVERWRITE & WIPE ANY DATA ON TARGET IMAGE FILE PARTITION)

- 1- Mount the target .img / .bin image file as a loop device: `losetup -fP <target image file>`
- 2- erase filesystem, raid or partition-table signatures (magic strings) from the specified **device PARTITION** USING: `wipefs --force -a <target loop device partition>`
- 3- Format the target loop device partition with desired fs-type. USING : mkfs command

[EXT4]: `mkfs.ext4 -F -L <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>`

[BTRFS]: `mkfs.btrfs -f -L <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>`

[FAT32]: `mkfs.fat -F 32 -n <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>`

[exFAT]: `mkfs.exfat -n <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>`

[NTFS]:

`mkfs.ntfs --quick --force --label <LABEL-OF-PARTITION> <TARGET-LOOP-DEV-PARTITION>`

- 4- Detach the loop device associated with the target image file. USING:
`losetup -d <loop dev>`

OVERVIEW TO PARTITIONATE A .IMG / .BIN (MULTIPLE PARTITIONS)

- 1- To create / modify / delete partitions in a .img / .bin file: use `cgdisk` command
`cgdisk <target-image-file>`